

Language Detection Model

Model Developed by

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GitHub: <https://github.com/1umairali/models>

Summary:

Developed a machine learning model capable of automatically identifying the language of a given text input with high accuracy. The model was trained on multilingual datasets and implemented using Python, leveraging natural language processing (NLP) techniques and supervised learning algorithms.

Key Contributions:

- Built and trained a language detection model using Multinomial Naive Bayes and text vectorization (TF-IDF/CountVectorizer).
- Preprocessed large multilingual datasets including tokenization, lowercasing, and stopwords removal.
- Evaluated model performance using accuracy, confusion matrix, and cross-validation techniques.
- Deployed model in an interactive Jupyter Notebook environment with visual output for user testing.

```
In [1]: import numpy as np # Linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.preprocessing import LabelEncoder
```

```
In [2]: url = "https://raw.githubusercontent.com/1umairali/models/main/language_detection/language_detection_dataset.csv"
data = pd.read_csv(url)
data
```

Out[2]:

	Text	language
0	klement gottwaldi surnukeha palsameeriti ning ...	Estonian
1	sebes joseph pereira thomas på eng the jesuit...	Swedish
2	ถนนเจริญกรุง ถนนโรมัน thanon charoen krung l...	Thai
3	விசாகப்பட்டினம் தமிழ்ச்சங்கத்தை இந்துப் பத்திர...	Tamil
4	de spons behoort tot het geslacht haliclona en...	Dutch
...	...	...
21995	hors du terrain les années et sont des année...	French
21996	ใน พศ หลังจากที่ได้ตั้งประพาสแหลมมลายู ชาว อิน...	Thai
21997	con motivo de la celebración del septuagésimoq...	Spanish
21998	年月，當時還只有歲的她在美國出道，以mai-k名義推出首張英文《baby i like》，由...	Chinese
21999	aprilie sonda spațială messenger a nasa și-a ...	Romanian

22000 rows × 2 columns

```
In [3]: df = data.copy()
df.sample(5)
```

Out[3]:

	Text	language
21778	acanthopetalum subpatens är en mångfotingart s...	Swedish
10130	au fost mai multe discuții despre a o face pe ...	Romanian
12758	les arbitres du championnat de france sont ama...	French
17598	นนลนีย โอแกน มกราคม พศ ประเทศสหรัฐอเมริกา -...	Thai
17067	павлова г е фёдоров а с михаил васильевич ломо...	Russian

```
In [4]: # check null values
df.isnull().sum()
```

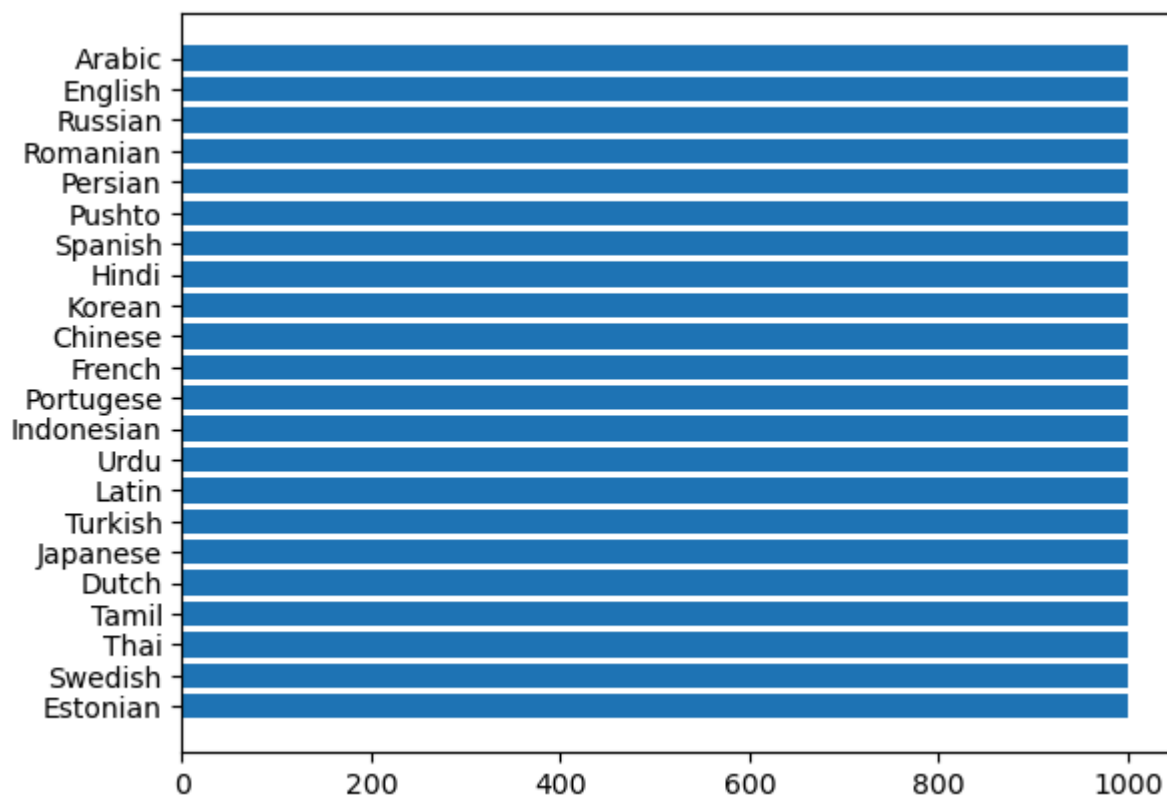
```
Out[4]: Text      0
language  0
dtype: int64
```

```
In [5]: # count values of each Language
df['language'].value_counts()
```

```
Out[5]: Estonian      1000
Swedish      1000
English      1000
Russian      1000
Romanian     1000
Persian      1000
Pushto       1000
Spanish      1000
Hindi        1000
Korean       1000
Chinese      1000
French       1000
Portugese    1000
Indonesian   1000
Urdu         1000
Latin        1000
Turkish      1000
Japanese     1000
Dutch        1000
Tamil        1000
Thai         1000
Arabic       1000
Name: language, dtype: int64
```

```
In [6]: # show language values in plot
plt.barh(df['language'].unique(), df['language'].value_counts(sort=False).values)
```

```
Out[6]: <BarContainer object of 22 artists>
```



Split data into Train and Test

```
In [7]: from sklearn.model_selection import train_test_split
```

```
In [8]: # assign data to train and test
x = df['Text']
y = df['language']

cv = CountVectorizer()
X = cv.fit_transform(x)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
In [9]: X_train
```

```
Out[9]: <17600x277720 sparse matrix of type '<class 'numpy.int64'>'
        with 732016 stored elements in Compressed Sparse Row format>
```

```
In [10]: X_test
```

```
Out[10]: <4400x277720 sparse matrix of type '<class 'numpy.int64'>'
        with 181582 stored elements in Compressed Sparse Row format>
```

```
In [11]: y_train
```

```
Out[11]: 5207      Thai
         4450      Thai
         7033    Swedish
         487      Tamil
        19537    Russian
         ...
        11964    Persian
        21575    Spanish
        5390    Turkish
         860     Pushto
        15795    Japanese
Name: language, Length: 17600, dtype: object
```

```
In [12]: y_test
```

```
Out[12]: 13035    Japanese
         3115     Russian
         8732     Latin
         7591    Pushto
          221     Hindi
         ...
        15215     Dutch
        12321    Chinese
        16643    English
        10475    Pushto
         9820     Dutch
Name: language, Length: 4400, dtype: object
```

Model

```
In [13]: from sklearn.naive_bayes import MultinomialNB
```

```
In [14]: model = MultinomialNB()
         model.fit(X_train,y_train)
```

```
Out[14]: ▾ MultinomialNB
         MultinomialNB()
```

```
In [15]: model.score(X_test,y_test)
```

```
Out[15]: 0.9529545454545455
```

Test Model

```
In [16]: user = input("Enter a Text & hit enter: ")
         data = cv.transform([user]).toarray()
         output = model.predict(data)
         print('Language is ',output)

         # Text sample for testing
         ###   Позже континенты воссоединились, образовав Паннотию   <----- Russian Language text
         ###   चलो बाहर जाकर खाना खाते हैं   <----- Hindi Language text
         ###   C'est un chien   <----- French Language text
```

Enter a Text & hit enter: Позже континенты воссоединились
Language is ['Russian']